

Day 5 (04/11/2025)

Operators -

It is special symbol used for particular operation between two operands.

$$a + b = c$$

$+$, $=$ are the operators

a , b , c are the operands

$$4 - 8 = -4$$

$-$ are the operators

4 , 8 , -4 are the operands

Types of operators in python.

Arithmetic Operators - Arithmetic operators are used to performing mathematical operations like addition, subtraction, multiplication, and division.

$+$, $-$, $*$, $/$, $\%$, $//$, $**$

Addition -

$+$ is the addition operator. It is used to add 2 values.

$$a = 2$$

$$b = 3$$

$$add = a + b$$

$$\text{print}(add) \quad \text{o/p} \Rightarrow 5$$

Subtraction -

$-$ is the subtraction operator. It is used to subtract the second value from the first value.

$$a = 2$$

$$b = 3$$

$$sub = a - b$$

$$\text{print}(sub) \quad \text{o/p} \Rightarrow -1$$

Logical Operators - Logic operators perform logical AND, logical OR and logical NOT operations. It is used to combine conditional statements.

a	b	AND AND	OR	NOR(NOT+OR)	XOR	XNOR	NAND
0	0	0	0	1	0	1	1
1	0	0	1	0	1	0	1
0	1	0	1	0	1	0	1
1	1	1	1	0	0	1	0

ex- age = 24
country = "india"
age >= 18 and country == "india" o/p
True

② ~~a~~ a = False
not(a) o/p
True

Bitwise Operator - Bitwise operators are used to perform bitwise calculations on integers. The integers are first converted into binary & then operations are performed on each bit (or) corresponding pair of bits, hence the name bitwise operators. The result is then returned in decimal format.

&, |, ~, >>, <<, ^

Bitwise AND (&) -

Take two bit values X and Y, where $x = 7 = (111)_2$ and $y = 4 = (100)_2$. Take Bitwise and of both x & y

$x = 7 \rightarrow 111$
 $y = 4 \rightarrow 100$
 $\underline{100} \rightarrow 4$
o/p ≥ 4

Bitwise OR-

ex - $X = 7 = (111)_2$, $Y = 4 = (100)_2$ - Take Bitwise OR of both X, Y

$$X = 7$$

$$Y = 4$$

print ($X | Y$)

$$X = 7 \Rightarrow 111$$

$$Y = 4 \Rightarrow \begin{array}{r} 100 \\ 111 \\ \hline \end{array} \Rightarrow 7$$

$$o/p = 7$$

Bitwise NOT-

$$X = 5 = (101)_2$$

$$X = 5$$

print ($\sim X$)

$$\sim 1001 \Rightarrow 0110 \Rightarrow 6$$

$$\begin{array}{r} o/p \\ -6 \end{array}$$

Bitwise Right Shift - shifts the bits of the number to the right & fills 0 on voids left (fills 1 in the case of a negative number) as a result. Similar effect as of dividing the number with some power of two.

$$\begin{aligned} \text{Ex-1} \Rightarrow a = -10 &= 0000\ 1010 \\ a \gg 1 &= 0000\ 0101 \Rightarrow 5 \end{aligned}$$

$$\begin{aligned} \text{Ex-2} \Rightarrow a = -10 &= 1111\ 0110 \\ a \gg 1 &= 1111\ 1011 \Rightarrow -5 \end{aligned}$$

Bitwise Left Shift - shifts the bits of the number to the left and fills 0 on voids right as a result. Similar effect as of multiplying the number with some power of two.

$$\begin{aligned} \text{Ex-1} \Rightarrow a = 5 &= 0000\ 0101 \\ a \ll 1 &= 0000\ 1010 = 10 \\ a \ll 2 &= 0001\ 0100 = 20 \end{aligned}$$

$$\text{Ex-2} \Rightarrow b = -10 = 1111\ 0110$$

$$b \ll 1 = 1110\ 1100 = -20$$

$$b \ll 2 = 1101\ 1000 = -40$$

Assignment Operator - These operators are used to assign values to the variables.

$$= \quad +=$$

$$-= \quad *=$$

$$/= \quad \>> \quad \% =$$

Addition Assignment -

$$a = 3$$

$$b = 5$$

$$a += b \Rightarrow a = a + b \Rightarrow 8$$

$$\begin{array}{r} o/p \\ 8 \end{array}$$

print (a)

Subtraction Assignment -

```
a=3
b=5
a-=b
print(a)
```

```
a=a-b
=3-5
=-2
o/p=>-2
```

Multiplication Assignment -

```
a=3
b=5
a*=b
print(a)
```

```
a=a*b
=3*5
o/p=>15
```

Division Assignment -

```
a=3
b=5
a/=b
print(a)
```

```
a=a/b
=3/5
o/p=>0.6
```

Modulus Assignment -

```
a=3
b=5
a%=b
print(a)
```

```
a=a%b
=3%5
o/p=>3
```

Floor division Assignment -

```
a=3
b=5
a//=b
print(a)
```

```
a=a//b
=3//5
o/p=>0
```

Exponential Assignment -

```
a=3
b=5
a**=b
print(a)
```

```
a=a**b
=3**5
o/p=>243
```

Membership Operator - In and NotIn are the membership operator that are used to test whether a value (or) variable is in a sequence.

in $\rightarrow$ True if value is found in the sequence.
not in $\rightarrow$ True if value is not found in the sequence.

```
grp = [1, 2, 5, 6, 7, 8, 3]
2 in grp - o/p=> True
```

```
grp = [1, 2, 5, 6, 7, 8, 3]
9 not in grp o/p=> True.
```


Identity Operator - `is` and `is not` are the identity operators both are used to check if two values are located on the same part of the memory. Two variables that are equal do not imply that they are identical.

`is` True if the operands are identical

`is not` True if the operands are not identical.

IS - `IS` operator checks if two variables point to the same object (same memory location),

```
a = [1, 2, 3]
```

```
b = [1, 2, 3]      o/p => False
```

```
c = a
```

```
print (a is c)
```

IS NOT - The opposite of `is`. It checks if two variables point to different objects.

Ex -

```
a = [1, 2, 3]
```

```
b = [1, 2, 3]
```

```
o/p => True.
```

```
c = a
```

```
print (a is not c)
```